Algorithmic Prediction of Trade and Value of Monetary and Currency-Based Assets:
A Final Report on Macro Stress, Safe-Haven Behavior, and Predictive Modeling

Nathan Dennis Dhruv Kartik Sebastian Vaskes Pimentel
CompSci 390B

Final Project Report – May 2026

Abstract

Financial stress changes how assets co-move, so average returns can hide the exact behavior that matters most for portfolio risk management. Using 4,150 daily observations from 2014-10-17 through 2026-02-25, we study whether macro-financial stress indicators help explain and predict the behavior of Gold, SPY, and Bitcoin under elevated stress regimes. Our approach combines regime-conditioned statistical tests, event studies, volatility and correlation analysis, chronological predictive modeling, threshold robustness, walk-forward validation, and SHAP interpretability. The final evidence supports Gold as a conditional safe haven: Gold defends most clearly during volatility-driven stress and the COVID crash, but not during every macro shock, while Bitcoin does not behave like "digital gold" under the same tests. XGBoost improves next-day Gold-versus-SPY classification to 0.6835 ROC–AUC, but persistence and Ridge remain difficult to beat for slow-moving volatility targets, and stress-regime forecasting remains limited by imbalance and regime drift.

1 Introduction and Motivation

Financial stress frequently propagates across asset classes. A widening high-yield spread, a spike in the VIX, or a broad deterioration in a financial stress index can coincide with equity drawdowns, changing correlations, liquidity pressure, and shifts into defensive assets. The practical question is not only whether an asset has a high average return. For risk management, the sharper question is whether an asset behaves defensively exactly when risky assets are under pressure.

This project studies that question for three assets: Gold, SPY as a proxy for U.S. equities, and Bitcoin. Gold is traditionally described as a safe-haven asset, while Bitcoin is sometimes described as "digital gold." Those labels are only useful if they survive common definitions of stress, event-window checks, and predictive modeling. We therefore frame the project around the following research questions:

- **RQ1: Stress and volatility.** Are volatility and market-risk measures systematically higher during macro-financial stress?
- **RQ2: Gold as a safe haven.** Does Gold behave defensively relative to SPY during stress, and is the result robust to stress definitions and thresholds?
- **RQ3: Bitcoin comparison.** Does Bitcoin behave like Gold under the same stress definitions, or does it behave more like a high-beta risk asset?
- **RQ4: Predictive modeling.** Can lagged macro-financial indicators and market-state variables improve forecasting beyond simple baselines?

Our final answer is deliberately conditional. Gold does not dominate in every crisis. It is strongest in volatility-driven and liquidity-like stress, especially the 2020 COVID crash; it is weaker in the 2022 inflation and rate-hike environment. Bitcoin fails the same safe-haven tests because its beta to SPY rises in stress, its volatility is persistently high, and it shares more left-tail risk with equities than Gold does.

2 Related Work

Safe-haven analysis usually distinguishes between assets that hedge risk on average and assets that protect specifically during market stress. Baur and Lucey define safe-haven behavior as low or negative co-movement with risky assets during extreme market conditions, not merely a positive long-run return [1]. That distinction motivates our use of stress-conditioned returns, rolling correlations, and event-window drawdowns.

The Bitcoin comparison is also motivated by mixed prior evidence. Bouri, Azzi, and Dyhrberg study Bitcoin's return-volatility behavior around early crashes and highlight that Bitcoin can behave differently from traditional safe-haven assets [2]. Corbet et al. show that cryptocurrencies have dynamic relationships with other financial assets and do not always provide stable diversification benefits [3]. Our contribution is to evaluate Gold, SPY, and Bitcoin under exactly the same stress definitions rather than treating the assets separately.

On the forecasting side, financial volatility is known to be persistent and regime-dependent. This makes simple persistence and regularized linear models hard baselines to beat. We still test nonlinear models because crisis relationships can be nonlinear and interaction-heavy. XGBoost is a strong tree-boosting method for tabular data [5]; SHAP provides a principled way to attribute tree-model predictions back to individual features [6]. Finally, because stress-regime prediction can be imbalanced, we evaluate imbalance-aware methods such as class weighting, threshold tuning, and SMOTE [7].

3 Data and Preprocessing

3.1 Dataset

We use the Kaggle dataset *Algorithmic Trading, Macro Stress, and Asset Regimes*, which contains 4,150 daily observations from 2014-10-17 through 2026-02-25 [4]. The asset variables include Gold, U.S. equities through `Equities_US` (used as SPY), and Bitcoin. The macro-financial variables include `Volatility_Index`, `High_Yield_Spread`, `Financial_Stress_Index`, `Yield_Curve_Spread`, `Stock_Bond_Corr_90d`, `SPY_Drawdown`, RSI features, and 30-day rolling volatility measures.

We transform prices to daily log returns,

$$r_{i,t} = \log(P_{i,t}) - \log(P_{i,t-1}),$$

for asset $i \in \{\mathrm{Gold}, \mathrm{SPY}, \mathrm{BTC}\}$. Stationarity checks in the earlier milestones showed that price levels are non-stationary while log returns are closer to stationary, so all return models use transformed returns rather than raw price levels.

3.2 Regime construction

We define stress regimes from high quantiles of three macro-stress proxies. The baseline threshold is the 66th percentile, which is high enough to represent elevated stress but still leaves enough observations for statistical comparison. For a stress proxy $X^{(j)}$, the stress flag is

$$S_t^{(j)}(q) = \mathbf{1}\left\{X_t^{(j)} \geq Q_q(X^{(j)})\right\},$$

where q is the chosen quantile. We use three single-proxy regimes and two composite regimes: `stress_hys`, `stress_fsi`, `stress_vix`, `stress_any`, and `stress_all`. Table 1 summarizes the baseline definitions.

Table 1: Baseline stress-regime definitions at the 66th percentile. The composite regimes reduce dependence on any single stress proxy.

Regime	Threshold / rule	Share of days
`stress_hys`	HYS ≥ 4.400	34.2%
`stress_fsi`	FSI ≥ -0.172	34.1%
`stress_vix`	VIX ≥ 19.060	34.0%
`stress_any`	At least one flag high	56.3%
`stress_all`	All three flags high	14.3%

3.3 Feature engineering and leakage control

The final supervised feature matrix contains 55 predictors. These include contemporaneous macro-financial levels, 30-day rolling volatilities, 1-, 3-, and 5-day lags of those variables, lagged returns for Gold, SPY, and Bitcoin, and contemporaneous stress indicators such as `stress_any_t`. Rolling calculations create unavoidable warm-up missingness, and forward targets drop the final rows. We therefore construct forward targets first, align features by timestamp, and drop only rows unavailable at prediction time.

All train/test splits are chronological. There is no random shuffling. Targets are shifted forward before splitting, so every supervised task uses only information available at time t to predict outcomes at $t+1$ or $t+5$. For the main safe-haven classifier, the training window runs from 2014-11-21 through 2023-11-25, and the held-out test window runs from 2023-11-26 through 2026-02-24. The test set has 822 rows, and Gold beats SPY on 38.3% of those days.

4 Methods

4.1 Descriptive safe-haven tests

The central descriptive statistic is the Gold–SPY daily log-return spread, $r_{Gold,t} - r_{SPY,t}$. For each regime, we compare the mean spread on high-stress days with the mean spread on low-stress days. A positive widening means Gold becomes more attractive relative to SPY during stress. We compute bootstrap 95% confidence intervals for the high-minus-low widening.

We also run a decile-resolved version of the test. Instead of comparing only high vs. low stress, each stress proxy is binned into deciles, and we compute the Spearman rank correlation between stress decile and the Gold–SPY spread. This directly tests whether safe-haven behavior strengthens monotonically as stress intensifies.

4.2 Volatility, correlation, and event windows

A safe haven should not merely have a positive average return. It should be less explosive than risky assets and should decouple from equities during stress. We therefore compare 30-day annualized volatility in calm and stress periods and examine 60-day rolling correlations between Gold and SPY and between Gold and Bitcoin. We also evaluate three event windows: the 2020 COVID crash, the 2022 inflation/rate-hike episode, and the 2023 banking-stress episode.

4.3 Predictive models

We evaluate three modeling families. **Model E** predicts whether Gold will outperform SPY the next day:

$$y_t^E = \mathbf{1}\{r_{Gold,t+1} > r_{SPY,t+1}\}.$$

The main models are a majority baseline, a VIX-threshold rule, class-balanced Logistic Regression, XGBoost, LightGBM, CatBoost, Random Forest, an MLP, and a stacking ensemble. We report accuracy, balanced accuracy, F1, ROC–AUC, Brier score, and log loss.

Model F forecasts 30-day rolling volatility for SPY, Bitcoin, and VIX at horizons $h = 1$ and $h = 5$. We compare persistence, Ridge regression, and XGBoost, reporting RMSE, MAE, and R^2.

Model G forecasts stress five trading days ahead. The original high-yield-only target was extremely imbalanced in the held-out window, so the final robustness block also evaluates a broader `stress_any` target with PR–AUC as the main metric. We test persistence, class-balanced Logistic Regression, XGBoost, `scale_pos_weight`, SMOTE, and validation-threshold tuning.

4.4 Interpretability and validation

For Model E, we use SHAP values from the fitted XGBoost model to interpret predictions on the chronological test window. We compare mean absolute SHAP rankings with gain importance and permutation importance, because tree gain can overstate some features when correlated predictors split similar signal. We also group features into economic themes: cross-asset momentum, macro-stress block, RSI technicals, volatility levels, and other macro variables.

Finally, we replace the single 80/20 split with expanding-window walk-forward validation. For each test year from 2019 through 2026, the model trains on all earlier observations and tests on that year's observations. This tests whether the Model E signal is concentrated in one crisis period or survives across changing regimes.

5 Safe-Haven Evidence

5.1 Return evidence by stress regime

Table 2 summarizes the strongest return evidence from the baseline 66th-percentile regime definitions. The clearest case is `stress_vix`: Gold's mean daily log return is positive, SPY's is negative, and the Gold–SPY spread widens by 0.00247. Composite stress also supports Gold relative to SPY: `stress_any` and `stress_all` have positive spread widening with bootstrap lower bounds above zero. The weaker result is `stress_fsi`, where Gold does not beat SPY on average.

The decile analysis sharpens this conclusion. The Spearman correlation between VIX decile and the Gold–SPY spread is positive and significant ($\rho = +0.0859$, $p < 0.001$), while the high-yield-spread decile correlation is small and negative ($\rho = -0.0257$, $p = 0.0979$), and the financial-stress-index decile correlation is also small and negative ($\rho = -0.0305$, $p = 0.0493$). In plain terms, Gold's defensive spread rises most consistently with volatility-style stress, not with every credit or financial-stress proxy.

5.2 Threshold robustness

Milestone 2 used the 66th percentile to define stress. To test whether the headline result depends on that threshold, we repeat the Gold–SPY widening test at the 66th, 75th, and 90th percentiles. Table 3 shows that the VIX-based result survives all three thresholds. The `stress_any` result survives at the 66th and 75th percentiles but not at the 90th percentile. HYS and FSI stress do not produce significant widening at any threshold.

5.3 Volatility and correlation

Stress raises volatility, but it does not raise all volatility equally. Under `stress_any`, Gold's annualized 30-day volatility rises from 0.1040 to 0.1298, a 24.8% increase. SPY rises from 0.0853 to 0.1521, a 78.3% increase. Bitcoin stays around 0.61 in both states, which means it is not stress-insensitive in a defensive sense; it is simply always highly volatile. Welch tests find Gold's and SPY's volatility increases statistically significant ($p < 0.001$), while Bitcoin's change is not significant ($p = 0.8592$).

Rolling correlations give a mixed but useful check. Gold–SPY correlation falls under several definitions: from 0.016 to -0.003 under `stress_hys`, from 0.045 to -0.059 under `stress_fsi`, and from 0.034 to -0.010 under `stress_any`. However, under the VIX definition it rises from -0.032 to 0.089. This means Gold sometimes decouples from equities during stress, but the correlation result is not universal. Gold–Bitcoin correlation generally rises in stress, including from 0.055 to 0.110 under `stress_any` and from 0.061 to 0.236 under `stress_all`, which weak-

Table 2: Mean daily log returns by baseline stress regime. Δ is the widening of the Gold–SPY spread from low stress to high stress. A positive CI lower bound supports a safe-haven interpretation relative to SPY.

Regime	Gold high	SPY high	BTC high	Δ(Gold–SPY)	CI lower	Interpretation
`stress_vix`	0.00068	-0.00092	-0.00008	0.00247	0.00157	Strong Gold-vs-SPY support
`stress_any`	0.00028	-0.00011	0.00008	0.00096	0.00029	Moderate Gold-vs-SPY support
`stress_all`	0.00095	-0.00067	0.00025	0.00191	0.00040	Strong in joint stress states
`stress_fsi`	0.00036	0.00046	-0.00003	-0.00012	-0.00102	No Gold-vs-SPY advantage

Table 3: Threshold robustness of the Gold–SPY widening test. Values are high-minus-low widening in daily log return spread. Stars indicate bootstrap 95% CI excludes zero.

q	Any	FSI	HYS	VIX
0.66	0.0010*	-0.0001	0.0004	0.0025*
0.75	0.0009*	-0.0001	0.0000	0.0023*
0.90	0.0011	-0.0000	0.0006	0.0025*

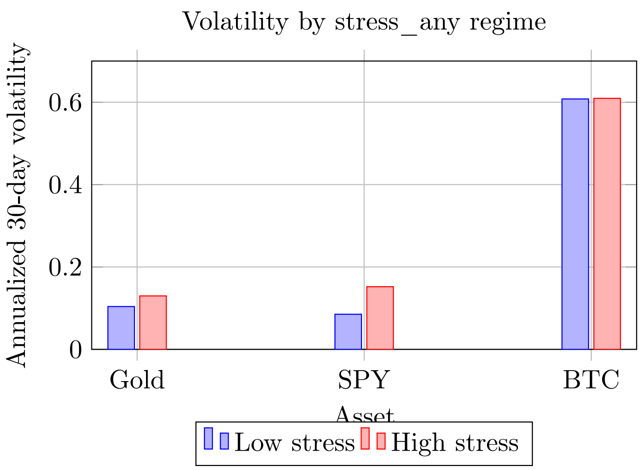

Figure 1: Gold volatility rises in stress but remains much lower than Bitcoin and expands much less than SPY. Bitcoin behaves like a persistently high-volatility asset.

ens the "digital gold" narrative.

5.4 Event windows

Event studies show why the final conclusion should be conditional rather than absolute. Figure 2 compares cumulative log returns during three major stress windows. In the COVID crash window, Gold is positive while SPY and Bitcoin are negative. In 2022, Gold loses value and underperforms SPY, although it falls far less than Bitcoin. In 2023 banking stress, Gold rises and beats SPY, but Bitcoin rises even more.

Peak-to-trough drawdowns tell the same story. During the COVID crash, Gold's maximum drawdown inside the window is -0.1338, compared with -0.4112 for SPY and -0.7131 for Bitcoin. During 2022 inflation/rates, the drawdowns are -0.1434 for Gold, -0.1825 for SPY, and -0.5255 for Bitcoin. During 2023 banking stress, the drawdowns are smaller for all three assets: -0.0293 for Gold, -0.0484 for SPY, and -0.1582 for Bitcoin.

5.5 Bitcoin is not digital Gold in this sample

The direct Bitcoin comparison is one of the strongest final findings. In calm periods, Gold's beta to SPY is +0.1569; during stress it falls to +0.0199. That is the desired decoupling pattern. Bitcoin's beta moves the opposite way: from +0.6657 in calm periods to +0.8617 during stress. Bitcoin also has more left-tail overlap with SPY. Of Gold's 5% worst days, 13.5% also fall in SPY's 5% worst days; for Bitcoin the overlap is 16.3%. Both are above the 5% random baseline, but Bitcoin is more aligned with equity tail risk.

6 Modeling Results

6.1 Milestone 1 baselines

The earliest modeling results established two baselines. Next-day Gold return prediction was weak: Ridge regression did not meaningfully improve on a zero-return baseline, with RMSE 0.010764 compared with 0.010732 for the zero baseline. This remains an important negative result because short-horizon returns are noisy even when medium-lag Granger tests suggest some directional relationship from equity returns to Gold returns.

Volatility forecasting was more successful. For 5-day-ahead SPY 30-day rolling volatility, Ridge improved on persistence in both RMSE and R^2 in Milestone 1. The later full Model F results confirm that this was not accidental: volatility is sticky enough that persistence and Ridge are very strong baselines.

6.2 Model E: predicting whether Gold beats SPY tomorrow

Model E is the strongest predictive task. Table 4 shows the final model comparison on the same chronological 80/20 split. XGBoost retains the best ROC–AUC at 0.6835, while the stacking ensemble is close at 0.6800. CatBoost and Random Forest deliver stronger balanced

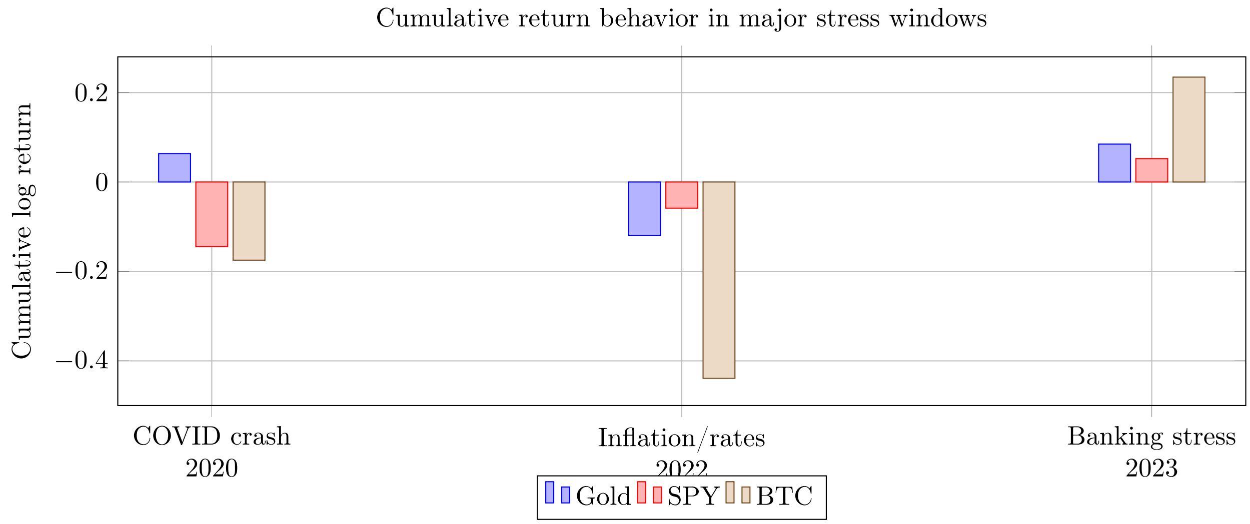

Figure 2: Gold protects capital most clearly in the COVID crash, is weak during the 2022 inflation/rate shock, and is positive but not dominant during the 2023 banking-stress window.

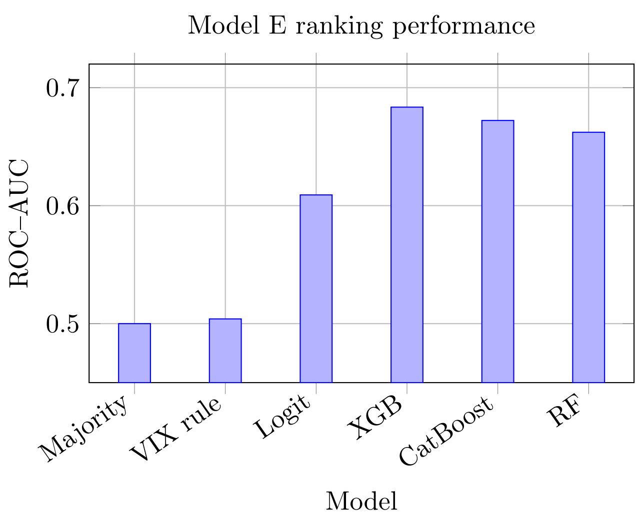

Figure 3: Nonlinear models improve the Gold-versus-SPY ranking task, with XGBoost leading by ROC–AUC.

accuracy than XGBoost, which matters if the goal is balanced classification rather than ranking. The MLP performs poorly, with high Brier score and log loss, indicating overconfident miscalibration.

The comparison to baselines is also important. A majority classifier has ROC–AUC 0.5000 by construction. A simple VIX-median rule reaches only 0.5040. Logistic Regression improves to 0.6091. XGBoost's 0.6835 therefore represents a meaningful ranking improvement, even though the F1 score shows many missed positive days.

6.3 Model F: volatility forecasting

Table 5 reports selected Model F results. The dominant pattern is that volatility is persistent. Ridge is best for SPY 30-day volatility at both $h = 1$ and $h = 5$. For Bitcoin volatility at $h = 5$, persistence and Ridge are close, and XGBoost is worse. For VIX at $h = 5$, XGBoost fails badly with negative out-of-sample R^2. This is not a failure of machine learning in general; it shows that model complexity must match the target. For slow-moving volatility, flexible nonlinear models can overfit spikes and lose to simple baselines.

6.4 Model G: stress-regime forecasting

The first stress-regime forecast targeted high-yield stress five days ahead and exposed a severe imbalance problem: the held-out test window had only about 0.7% positives. Accuracy was therefore misleading. The majority baseline achieved very high accuracy but zero F1 and balanced accuracy 0.5000. A persistence rule improved balanced accuracy, while Logistic Regression ranked rare positives well but needed better thresholding.

The final imbalance-aware block evaluates the broader `stress_any` five-day target and reports PR–AUC, which is more informative when positive rates shift. Table 6 shows that class-balanced Logistic Regression achieves the best PR–AUC at 0.7095 against a random baseline/base rate of 0.2433. XGBoost variants cluster around PR–AUC 0.65. The validation-tuned threshold of 0.922 improves validation F1 but generalizes poorly, which is evidence of regime drift between training, validation, and test periods.

7 Interpretability and Robustness

7.1 What drives the safe-haven classifier?

SHAP analysis shows that Model E is not simply a macro-stress indicator in disguise. Its largest

Table 4: Model E results on the chronological test window. The target is whether Gold beats SPY tomorrow. XGBoost remains the best model by ROC–AUC, while Random Forest has the strongest balanced accuracy.

Model	Accuracy	Bal. acc.	F1	ROC–AUC	Brier	Log loss
XGBoost	0.6460	0.5922	0.4393	**0.6835**	**0.2210**	0.6317
Stacking (xgb+lgb+logit)	0.6058	0.4995	0.0795	0.6800	**0.2210**	0.6314
CatBoost	0.6399	0.6179	0.5272	0.6722	0.2219	**0.6297**
LightGBM	0.6363	0.5993	0.4818	0.6687	0.2387	0.6835
Random Forest	**0.6472**	**0.6467**	**0.5833**	0.6622	0.2267	0.6455
Logistic (balanced)	0.5766	0.5883	0.5360	0.6091	0.2459	0.6945
MLP (2x64)	0.5292	0.5450	0.4994	0.5673	0.4377	3.6077

Table 5: Selected Model F results. Persistence and Ridge dominate XGBoost for slow-moving volatility targets. Bold marks the best RMSE within each target-horizon block.

Target	Horizon/model	RMSE	MAE	R^2
SPY 30d vol, $h = 1$	Persistence	0.0099	0.0035	0.9786
SPY 30d vol, $h = 1$	Ridge	**0.0097**	0.0043	**0.9796**
SPY 30d vol, $h = 1$	XGBoost	0.0164	0.0068	0.9421
SPY 30d vol, $h = 5$	Persistence	0.0248	**0.0123**	0.8670
SPY 30d vol, $h = 5$	Ridge	**0.0210**	0.0125	**0.9046**
SPY 30d vol, $h = 5$	XGBoost	0.0246	0.0151	0.8699
BTC 30d vol, $h = 5$	Persistence	**0.0608**	**0.0401**	**0.8074**
BTC 30d vol, $h = 5$	Ridge	0.0612	0.0460	0.8045
BTC 30d vol, $h = 5$	XGBoost	0.0755	0.0522	0.7029
VIX, $h = 5$	Persistence	**3.2783**	**1.8604**	**0.4950**
VIX, $h = 5$	Ridge	3.2904	2.0936	0.4912
VIX, $h = 5$	XGBoost	5.5236	4.4839	-0.4337

Table 6: Final Model G imbalance-aware results for five-day-ahead `stress_any`. PR–AUC is the headline metric because the positive rate shifts across time.

Model	Bal. acc.	F1	ROC–AUC	PR–AUC
Majority baseline	0.5000	0.3914	0.5000	0.2433
Regime persists	**0.7761**	**0.6633**	0.7761	0.5240
Logistic (balanced)	0.7376	0.6113	**0.8644**	**0.7095**
XGBoost default	0.7651	0.6234	0.8514	0.6429
XGB + weight	0.7530	0.6147	0.8539	0.6518
XGB + SMOTE	0.7352	0.5947	0.8559	0.6518
XGB + tuned threshold	0.6317	0.4261	0.8539	0.6518

contributions are short-horizon cross-asset momentum features: `spy_ret_lag3`, `spy_ret_lag1`, `gold_ret_lag1`, and `gold_ret_lag3`. The top macro-stress features include `High_Yield_Spread_lag5`, `Financial_Stress_Index_lag5`, and `Volatility_Index`, but the strongest block is lagged returns.

Grouping all 55 features into economic themes makes the interpretation clearer. Cross-asset momentum contributes 39.5% of total mean absolute SHAP attribution, the macro-stress block contributes 20.0%, RSI technicals 17.3%, volatility levels 14.3%, and other macro variables 9.0%. Thus the classifier is best described as a momentum-plus-stress model, not a pure macro-stress model.

The interpretability cross-check is useful. Spearman rank correlations between importance methods are modest: gain vs. SHAP is +0.204, gain vs. permutation is +0.389, and SHAP vs. permutation is +0.341. This means gain importance alone is not a reliable explanation. Several features with high gain importance fall under permutation, so the final interpretation relies on SHAP and economic grouping rather than gain alone.

The Logistic Regression comparison reinforces that XGBoost is finding a different signal. The Spearman correlation between XGBoost and Logistic mean absolute contribution rankings is -0.110 across all 55 features, and the top-15 feature overlap is only 4/15. XGBoost's lift is therefore not just a nonlinear version of the same linear ranking; it relies more heavily on lagged return interactions.

7.2 Does the model switch during stress?

Segmented SHAP analysis compares calm and stress test days. The macro-stress block contributes 20.9% of the signal on calm days but only 16.9% on stress days, while momentum and volatility levels rise. The interpretation is that when stress is already realized, return lags and volatility levels carry much of the regime information, so the explicit stress flag becomes less necessary.

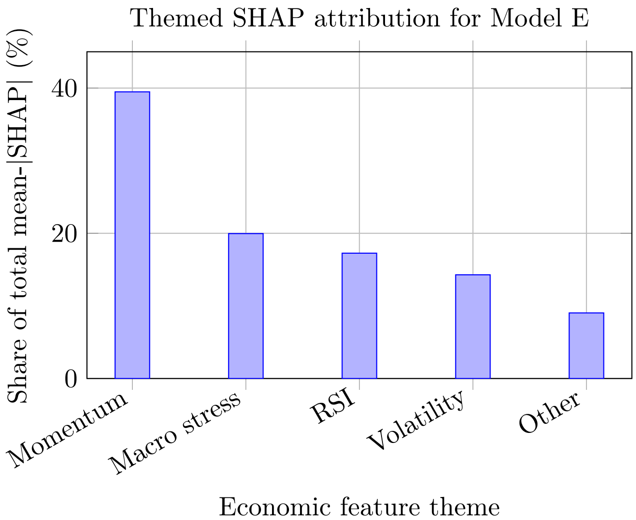

Figure 4: Most of XGBoost's signal comes from lagged cross-asset returns, followed by macro-stress variables and RSI features.

7.3 Walk-forward validation

Table 7 reports expanding-window validation for Model E. The mean per-year ROC–AUC is 0.6571 with standard deviation 0.0289, and all eight test folds clear 0.55 ROC–AUC. This lowers the concern that the 0.6835 single-split result is driven by one lucky train/test cut. The signal weakens in some years but stays consistently above chance.

7.4 Why the MLP failed

The MLP result is a useful negative finding. The Milestone 2-style MLP reached 100% training accuracy but only about 53% test accuracy, a 47-point generalization gap. Seven of the top ten SHAP-ranked features fail a Kolmogorov-Smirnov train/test distribution test at $p < 0.01$, including `High_Yield_Spread_lag5`, `Financial_Stress_Index_lag5`, and `GLD_RSI_14`. A fixed MLP with heavier regularization and early stopping improves calibration, cutting Brier score from roughly 0.44 to 0.24, but ROC–AUC remains near 0.51. The failure is therefore not only overfitting; it is also covariate shift.

8 Discussion

8.1 Main findings

Three conclusions integrate the full project. First, stress variables align strongly with volatility. High-yield spreads, the financial stress index, VIX, drawdowns, and realized volatility move together enough to define meaningful market states.

Second, Gold is a conditional safe haven. It behaves defensively relative to SPY under VIX stress, composite stress, and the COVID crash. It is far less volatile than Bitcoin and has a much smaller stress beta to equities. However, Gold is not a universal crisis asset: it underperforms SPY in the 2022 inflation/rate-hike event, and the correlation evidence depends on the regime definition.

Third, Bitcoin does not behave like Gold under the same tests. Its SPY beta rises in stress, its volatility is persistently high, and its left-tail overlap with SPY is larger than Gold's. Bitcoin can rally in some stress episodes, such as the 2023 banking-stress window, but that is not the same as stable safe-haven behavior.

On the modeling side, the strongest success is Model E. XGBoost, CatBoost, Random Forest, and the stacking ensemble show that nonlinear models can help rank days when Gold will beat SPY. The strongest negative result is Model F: for volatility targets, persistence and Ridge remain stronger and more stable than XGBoost. Model G shows that stress forecasting is sensitive to the target definition, class balance, and threshold choice.

8.2 Limitations and potential biases

Several limitations affect interpretation. **Regime sensitivity** remains important. VIX-based stress produces the most robust safe-haven result, while HYS and FSI stress do not. A different threshold or proxy can change the conclusion.

Full-sample threshold bias is also possible. The descriptive regime thresholds use quantiles computed over the full sample. That is acceptable for retrospective analysis but would leak future information in a live trading system. A real-time implementation should compute rolling thresholds using only data available at the time.

Non-stationarity and covariate shift are major constraints. The sample includes COVID, inflation/rate repricing, banking stress, and a changing crypto market. Relationships estimated before 2023 may not hold in 2024–2026. Walk-forward validation helps, but it cannot eliminate regime drift.

Event-window selection can bias the narrative. We selected economically meaningful stress windows, but different start and end dates could alter cumulative returns and drawdowns. Future work should define event windows algorithmically from stress indicators.

Asset and data limitations also matter. The dataset begins in 2014, which is enough to include modern Bitcoin behavior but not enough to evaluate Gold across many historical cycles. The stress variables are U.S.-centric, so results may not generalize to all global investors. We also do not model transaction costs, bid-ask spreads, taxes, or portfolio constraints.

Multiple testing is another concern. Many regimes, thresholds, horizons, and models are compared. Bootstrap intervals, chronological testing, and walk-forward validation reduce this risk, but they do not fully replace a

Table 7: Expanding-window walk-forward validation for Model E. Each year is tested using all prior observations for training.

Test year	Train n	Test n	Positive rate	ROC–AUC	Bal. acc.	Brier
2019	1502	365	0.3096	0.6459	0.5165	0.2114
2020	1867	366	0.3634	0.6214	0.5521	0.2341
2021	2233	365	0.3260	0.6158	0.5655	0.2303
2022	2598	365	0.3589	0.6696	0.5173	0.2370
2023	2963	365	0.3123	0.6608	0.5918	0.2109
2024	3328	366	0.3716	0.6600	0.5533	0.2288
2025	3694	365	0.3836	**0.7005**	0.5939	0.2101
2026	4059	55	0.4727	0.6830	**0.6273**	0.2314
Mean	–	–	–	0.6571	0.5641	0.2243

pre-registered design or an entirely independent holdout sample.

9 Future Work

The next step is to turn the descriptive and classification results into a real portfolio rule. For example, a strategy could tilt toward Gold only when a stress indicator is high and Model E predicts Gold outperformance. That rule should be tested against static 60/40, SPY-only, and Gold/SPY risk-parity benchmarks with transaction costs and turnover constraints.

A second extension is a real-time regime framework. Instead of full-sample quantiles, future work should use rolling or expanding quantile thresholds and evaluate how often the safe-haven claim survives with only past data. This would make threshold robustness directly deployable.

A third extension is shock-type modeling. The 2022 result suggests that inflation and real-rate shocks differ from liquidity or equity-volatility shocks. Adding real rates, dollar strength, inflation surprises, and central-bank variables could separate volatility-driven stress from rate-driven stress.

A fourth extension is stronger calibration. Model E ranks well but still misses many positive days at a 0.5 threshold. Isotonic calibration, Platt scaling, cost-sensitive losses, and threshold selection based on trading utility rather than F1 could improve practical decision-making.

Finally, Model G needs a more careful rare-event framework. PR–AUC should remain central, but future versions should add time-series cross-validation, Bayesian or conformal uncertainty, and explicit drift detection. The threshold that works in one period should not be assumed to generalize.

10 Conclusion

This project began with a broad question about whether macro-financial stress indicators can predict asset behavior. The final answer is nuanced. Stress indicators do contain useful information, but the usefulness depends on the task. They help describe volatility regimes and contribute to predicting whether Gold beats SPY, but they do not make short-horizon Gold returns easy to forecast and they do not beat simple baselines for sticky volatility targets.

The main substantive conclusion is that Gold is a conditional safe haven. It defends most clearly in volatility-driven stress and in the COVID crash, but it is not universally defensive across every stress definition or event window. Bitcoin does not pass the same safe-haven tests: it remains much more volatile, becomes more equity-sensitive in stress, and shares more equity tail risk. The best modeling conclusion is similarly conditional: advanced nonlinear models help when the target is a directional, regime-sensitive classification problem, but simple baselines remain powerful when the target is persistent volatility.

Group Contributions

Dhruv Kartik implemented the empirical pipeline, including regime construction, conditional return and volatility analysis, event studies, nonlinear modeling, SHAP analysis, threshold robustness, and walk-forward validation. Sebastian Vaskes Pimentel contributed to model design, training, evaluation, and comparisons across Logistic Regression, Ridge, XGBoost, LightGBM, CatBoost, Random Forest, MLP, and stacking models. Nathan Dennis led the written report structure, interpretation of statistical and modeling results, and integration of the milestones into the final standalone document. All members collaborated on the research questions, feature choices, safe-haven interpretation, limitations, and final conclusions.

References

[1] Dirk G. Baur and Brian M. Lucey. 2010. Is Gold a Safe Haven? International Evidence. *Journal of Banking &*

Finance, 34(8):1886–1898.

[2] Elie Bouri, Georges Azzi, and Anne Haubo Dyhrberg. 2017. On the Return-Volatility Relationship in the Bitcoin Market Around the Price Crash of 2013. *Economics: The Open-Access, Open-Assessment E-Journal*, 11(1):1–16.

[3] Shaen Corbet, Andrew Meegan, Charles Murphy, John Mutenga, and Charles Larkin. 2018. Exploring the Dynamic Relationships Between Cryptocurrencies and Other Financial Assets. *Economics Letters*, 165:28–34.

[4] Kanchana1990. 2024. *Algorithmic Trading, Macro Stress, and Asset Regimes Dataset*. Kaggle.

[5] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.

[6] Scott M. Lundberg and Su-In Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems*.

[7] Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. 2002. SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16:321–357.

[8] S. S. Anem, M. Amiruzzaman, and A. A. Bhuiyan. 2024. Studying Financial Data with Macroeconomic Factors Using Machine Learning. *Journal of Computing Sciences in Colleges*, 40(3):151–163.